

Exam Preparation Notes

Big Data Analytics & Business Intelligence

BE – Artificial Intelligence & Data Science

12 Units | All Important Topics | Formulas & Key Terms

BIG DATA ANALYTICS (BDA)

UNIT 1 – Introduction to Big Data and Analytics

1.1 Big Data – Definition & Evolution

Big Data refers to datasets so large and complex that traditional data-processing software is unable to deal with them. The term describes not just volume but a whole ecosystem of problems and solutions.

- **Definition (Gartner):** Data with high Volume, high Velocity, and high Variety that demands cost-effective, innovative forms of information processing.
- **Evolution:** Mainframes (1950s) → Relational DBs (1970s) → Data Warehouses (1990s) → Hadoop/NoSQL (2000s) → Cloud & Real-time (2010s+)

1.2 The 5 V's of Big Data (Characteristics)

V	Description	Real-world Example
Volume	Amount/size of data generated	Facebook: 4 PB/day; YouTube: 500 hours video/min
Velocity	Speed at which data is generated	Twitter: 6,000 tweets/sec; Stock ticks: microseconds
Variety	Different types and formats	Structured (SQL), Semi (JSON/XML), Unstructured (video)
Veracity	Accuracy and trustworthiness	Noisy sensor data, typos in form entries
Value	Business worth of the data	Customer insights → targeted campaigns → revenue

1.3 Challenges with Big Data

- **Storage:** Storing petabytes across distributed nodes reliably
- **Processing Speed:** Real-time or near-real-time analysis requirements
- **Data Quality (Veracity):** Inconsistent, incomplete, duplicate, or noisy data
- **Security & Privacy:** GDPR, HIPAA compliance; encrypting data at rest and in transit
- **Integration:** Combining heterogeneous sources (SQL, NoSQL, streaming, files)
- **Scalability:** Systems must scale horizontally as data grows
- **Skills Gap:** Shortage of data engineers, scientists, and architects

1.4 Traditional BI vs Big Data

Aspect	Traditional BI	Big Data Analytics
Data Type	Structured only (SQL)	Structured + Semi + Unstructured
Data Volume	GB to TB	TB to Petabytes (PB)
Processing	Batch (nightly ETL)	Batch + Streaming (real-time)
Storage	RDBMS, Data Warehouse	HDFS, NoSQL, Data Lakes, Cloud
Schema	Schema-on-write (rigid)	Schema-on-read (flexible)
Tools	SQL, SSRS, Crystal Reports	Hadoop, Spark, Hive, Kafka, Flink
Latency	Hours to days	Milliseconds to minutes
Cost	High (proprietary)	Low (open-source, commodity HW)

1.5 Big Data Ecosystem – Key Components

- **HDFS:** Hadoop Distributed File System — stores files split into 128 MB blocks across nodes
- **MapReduce:** Programming model for parallel processing — Map (split) → Shuffle → Reduce (aggregate)
- **YARN:** Yet Another Resource Negotiator — manages cluster resources and job scheduling

- **Apache Spark:** In-memory processing engine; 100× faster than MapReduce for iterative ML
- **Apache Hive:** SQL-like (HiveQL) query engine on top of Hadoop
- **Apache Pig:** High-level scripting (Pig Latin) for ETL on Hadoop
- **Apache Kafka:** Real-time distributed data streaming platform
- **HBase:** Column-family NoSQL database on HDFS (random read/write)
- **Zookeeper:** Coordination service — leader election, configuration, naming

1.6 Types of Analytics

Type	Question	Techniques	Example
Descriptive	What happened?	Aggregation, reports, dashboards	Monthly sales summary
Diagnostic	Why did it happen?	Drill-down, correlation, root cause	Why did Q3 sales drop?
Predictive	What will happen?	Regression, ML, forecasting	Next quarter revenue forecast
Prescriptive	What should we do?	Optimization, simulation, NLP	Optimal pricing strategy

1.7 Hadoop Architecture (Deep Dive)

- **NameNode (Master):** Manages HDFS metadata — file names, block locations, replication factor. Single point of failure (mitigated by HA NameNode).
- **DataNode (Slave):** Stores actual data blocks; sends heartbeat to NameNode every 3 seconds.
- **ResourceManager:** YARN master — accepts job submissions, allocates resources.
- **NodeManager:** YARN slave — manages containers on each node, reports to ResourceManager.
- **Replication:** Default replication factor = 3 (one local, one rack-local, one remote rack).

1.8 RapidMiner, Looker & Soft State

- **RapidMiner:** Visual drag-and-drop ML platform; 1500+ operators; no coding needed. Used for predictive analytics, text mining, image analytics.
- **Looker:** Google's BI & data exploration platform. Uses LookML (modelling language). Embeds in apps.
- **Soft State (BASE):** Distributed systems may temporarily hold inconsistent state.

- BASE = Basically Available, Soft state, Eventual Consistency
- Opposite of ACID (Atomicity, Consistency, Isolation, Durability)
- Eventual Consistency: given enough time, all replicas will converge

■ *Exam tip: V's of Big Data (5 V's), ACID vs BASE, Traditional BI vs Big Data table are very common exam questions.*

UNIT 2 – Basic Data Analytic Methods

2.1 Clustering – Overview

Clustering is an unsupervised machine learning technique that groups similar data points together into clusters based on their attributes — without using predefined labels.

- **Goal:** Maximize intra-cluster similarity (cohesion) + minimize inter-cluster similarity (separation)
- **Applications:** Customer segmentation, document grouping, anomaly detection, image compression
- **Types:** Partitional (K-Means), Hierarchical, Density-based (DBSCAN), Model-based (GMM)

2.2 K-Means Algorithm (Step-by-Step)

- **Input:** Dataset D, number of clusters K
- **Step 1:** Choose K initial centroids randomly from the data
- **Step 2: Assignment** — Assign each data point to the nearest centroid using Euclidean distance:
■ $d(x, c) = \sqrt{\sum (x_i - c_i)^2}$
- **Step 3: Update** — Recalculate each centroid as the mean of all points assigned to it
- **Step 4:** Repeat Steps 2–3 until centroids stop changing (convergence)
- **Output:** K clusters with their centroids
- **Limitations:** Must specify K; sensitive to outliers and initial centroid placement; assumes spherical clusters

2.3 Cluster Diagnostics & Validation

- **Elbow Method:** Plot WCSS (Within-Cluster Sum of Squares) vs K. Optimal K is at the 'elbow' point where gain decreases sharply.
■ $WCSS = \sum \sum ||x_i - c_k||^2$
- **Silhouette Score:** Measures how similar a point is to its own cluster vs other clusters. Range: -1 to +1 (higher is better).
■ $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$
■ $a(i)$ = avg distance to points in same cluster; $b(i)$ = avg distance to nearest other cluster
- **Davies-Bouldin Index:** Lower value = better separation (0 is perfect).
- **Dunn Index:** Higher value = compact, well-separated clusters.

2.4 Association Rules – Concepts

Association Rule Mining discovers interesting relationships (co-occurrences) between variables in large transaction databases. Classic use: Market Basket Analysis.

Metric	Formula	Meaning
Support	$\text{freq}(A \cup B) / N$	How often the itemset appears (frequency)
Confidence	$\text{Support}(A \cup B) / \text{Support}(A)$	How often the rule is correct
Lift	$\text{Confidence}(A \rightarrow B) / \text{Support}(B)$	Lift > 1: A and B are positively correlated
Conviction	$(1 - \text{Support}(B)) / (1 - \text{Confidence})$	Higher = stronger rule

■ **Rule:** {Bread} → {Butter} with Support=0.4, Confidence=0.8, Lift=1.6 means: 40% of transactions have both, 80% of bread buyers buy butter, and it's 1.6× more likely than by chance.

2.5 Apriori Algorithm (Step-by-Step)

- **Principle:** Anti-monotone property — if an itemset is infrequent, ALL its supersets are also infrequent (prune them).
- **Step 1:** Set minimum support threshold (min_sup) and confidence threshold (min_conf)
- **Step 2:** Scan database → find frequent 1-itemsets (L1)

- **Step 3:** Generate candidate 2-itemsets (C2) by joining L1 with itself
- **Step 4:** Prune C2 — remove any whose subsets are not in L1
- **Step 5:** Scan database → find frequent 2-itemsets (L2)
- **Step 6:** Repeat steps 3–5 until no more frequent itemsets
- **Step 7:** Generate association rules from frequent itemsets that meet min_conf

2.6 Linear Regression

Predicts a continuous output variable (dependent) from one or more input variables (independent).

- **Simple Linear:** $y = b_0 + b_1x$ (one predictor)
- **Multiple Linear:** $y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$
- **Cost Function:** $MSE = (1/n) \sum (y_i - \hat{y}_i)^2$ — minimized using Gradient Descent or OLS
- **Evaluation:** R^2 (variance explained), RMSE (prediction error), MAE
- **Assumptions:** Linearity, Independence, Homoscedasticity, Normality of residuals

2.7 Logistic Regression

Used for binary classification. Outputs probability between 0 and 1 using the sigmoid (logistic) function.

- **Sigmoid function:** $\sigma(z) = 1 / (1 + e^{-z})$ where $z = b_0 + b_1x_1 + \dots + b_nx_n$
- **Decision rule:** $P \geq 0.5 \rightarrow \text{Class 1}; P < 0.5 \rightarrow \text{Class 0}$
- **Cost Function:** Binary Cross-Entropy (Log Loss)
- **Use cases:** Spam detection, disease prediction, credit default, customer churn

2.8 Validation & Testing

- **Holdout / Train-Test Split:** Typically 70/30 or 80/20. Simple but high variance estimate.
- **K-Fold Cross Validation:** Data split into K equal folds. Model trained K times (each fold as test once). Final score = average of K scores.
- **Stratified K-Fold:** Maintains class proportions in each fold — recommended for imbalanced datasets.
- **Overfitting:** Low training error, high test error — model memorized training data.
- **Underfitting:** High training AND test error — model too simple.

■ Common K values: 5 or 10. Leave-One-Out (LOO) = K equals N (number of samples).

UNIT 3 – Predictive Analysis Process and R

3.1 R Language & RStudio GUI

R is a free, open-source statistical programming language and environment for statistical computing and graphics.

- **RStudio Layout:** Script Editor (top-left), Console (bottom-left), Environment/History (top-right), Files/Plots/Help (bottom-right)
- **Key Data Structures:** Vector, Matrix, Data Frame, List, Factor
- **Basic commands:** ls() — list variables; rm() — remove; str() — structure; summary() — statistics

3.2 Data Import & Export in R

Operation	Function	Example
Read CSV	read.csv()	df <- read.csv("data.csv", header=TRUE)
Read Excel	readxl::read_excel()	df <- read_excel("data.xlsx", sheet=1)
Read Table	read.table()	df <- read.table("data.txt", sep="\t")
Write CSV	write.csv()	write.csv(df, "output.csv", row.names=FALSE)
Write Excel	writexl::write_xlsx()	write_xlsx(df, "output.xlsx")
From Database	DBI::dbConnect()	con <- dbConnect(RSQLite::SQLite(), "db.sqlite")

3.3 Data Cleaning in R

- **Check Missing:** is.na(df) | colSums(is.na(df)) | sum(is.na(df))
- **Remove NAs:** df <- na.omit(df) | df <- df[complete.cases(df),]
- **Replace NAs:** df\$col[is.na(df\$col)] <- mean(df\$col, na.rm=TRUE)
- **Find Duplicates:** duplicated(df) | df[!duplicated(df),]
- **Outlier detection:** boxplot(df\$col) | IQR method: $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$
- **Change type:** as.numeric(), as.factor(), as.character(), as.Date()

3.4 Linear Regression in R

- model <- lm(Sales ~ Advertising + Price, data = df)
- summary(model) # Coefficients, R^2 , p-values, F-statistic
- predict(model, newdata = df_new) # Predict on new data
- plot(model) # 4 diagnostic plots: residuals, QQ, scale-location, leverage
- **Interpret R^2 :** 0.8 means 80% of variance in Y is explained by X

3.5 Clustering in R

- km <- kmeans(df, centers = 3, nstart = 25) # K-Means
- km\$cluster # Cluster assignments per row
- km\$centers # Centroid coordinates
- hc <- hclust(dist(df), method='ward.D2') # Hierarchical
- plot(hc) # Dendrogram visualization
- cutree(hc, k=3) # Cut dendrogram to get 3 clusters

3.6 Hypothesis Testing

Statistical hypothesis testing determines whether an observed effect in data is statistically significant or just due to chance.

- **H₀ (Null Hypothesis):** Default claim — no effect, no difference

- **H₁ (Alternate Hypothesis):** What we want to prove — some effect exists
- **p-value:** Probability of observing results at least as extreme as data IF H₀ is true.
 - $p < 0.05 \rightarrow$ Reject H₀ (result is statistically significant at 5% level)
 - $p \geq 0.05 \rightarrow$ Fail to reject H₀
- **t-test:** Compare means of two groups — `t.test(group1, group2)`
- **Chi-square test:** Independence of categorical variables — `chisq.test(table)`
- **ANOVA:** Compare means of 3+ groups — `aov(y ~ factor, data=df)`

3.7 MapReduce – Programming Model

MapReduce is a programming paradigm for processing large datasets in parallel across a distributed cluster.

- **Map Phase:** Each mapper processes a split of input data independently and emits (key, value) pairs.
- **Shuffle & Sort:** Framework groups all values with the same key together and sends to reducers.
- **Reduce Phase:** Each reducer aggregates values for a key and produces final output.
- **WordCount Example:** Map: ('hello',1),('world',1) $\rightarrow$ Reduce: ('hello',2),('world',1)

■ **MapReduce Fault Tolerance:** If a task fails, the framework automatically reruns it on another node.

3.8 Data Analytics Lifecycle (6 Phases)

Phase	Key Activities	Deliverable
1. Discovery	Understand business goals, data sources	Problem statement, data inventory
2. Data Preparation	ETL, cleaning, feature engineering, EDA	Clean, analysis-ready dataset
3. Model Planning	Select ML techniques, variables, tools	Modelling plan, feature list
4. Model Building	Train models, hyperparameter tuning, iteration	Trained model(s)
5. Evaluate Results	Test vs business success criteria, interpret	Model evaluation report
6. Operationalize	Deploy, monitor, maintain, retrain as needed	Production model, monitoring dashboard

3.9 Predictive Model Building Steps

1. **Define Problem:** Clearly state what to predict (target variable)
2. **Collect Data:** Gather historical data from relevant sources
3. **Preprocess:** Clean, normalize, encode categorical variables
4. **Feature Engineering:** Create new features, select most informative ones
5. **Split Data:** Train/validation/test sets
6. **Train Model:** Fit algorithm to training data
7. **Evaluate:** Measure performance on test set using appropriate metrics
8. **Tune:** Hyperparameter optimization (Grid Search, Random Search)
9. **Deploy:** Integrate into production pipeline or application

UNIT 4 – Advanced Predictive Analytics Algorithms and Python

4.1 Exploratory Data Analysis (EDA)

EDA is the critical first step in analyzing data. It uses statistical summaries and graphical representations to understand data structure, detect outliers, find patterns, and test assumptions before modeling.

- **Univariate EDA:** One variable at a time — mean, median, mode, std dev, skewness, kurtosis; histogram, box plot, KDE plot
- **Bivariate EDA:** Two variables — scatter plot, correlation matrix, heat map, bar chart by group
- **Multivariate EDA:** Multiple variables — pair plot (seaborn.pairplot), PCA biplot, parallel coordinates
- **Python tools:** pandas (describe(), info()), matplotlib (hist, scatter), seaborn (heatmap, pairplot), plotly (interactive)

4.2 Ensemble Methods – Overview

Ensemble learning combines predictions from multiple models to produce better accuracy and robustness than any single model — 'Wisdom of the Crowd' principle.

Method	Idea	Reduces	Example
Bagging	Parallel models on random subsets	Variance	Random Forest
Boosting	Sequential models, focus on errors	Bias	AdaBoost, XGBoost
Stacking	Meta-model combines predictions	Both	Blend of RF + GBM + LR

4.3 Bagging (Bootstrap Aggregating)

- **How it works:** Train B models, each on a different bootstrap sample (random sample WITH replacement) from training data.
- **Aggregation:** Classification → majority vote; Regression → average of predictions
- **Effect:** Reduces variance (overfitting) without increasing bias
- **Out-of-bag (OOB) error:** ~37% of samples not used in each bag → free validation set

4.4 Boosting

- **How it works:** Models trained sequentially. Each new model focuses more on samples misclassified by previous models (by increasing their weights).
- **Effect:** Reduces bias (underfitting) — turns weak learners into a strong learner
- **Risk:** Can overfit if too many estimators or noisy data

4.5 AdaBoost (Adaptive Boosting)

- **Step 1:** Initialize equal weights: $w_i = 1/N$ for all samples
- **Step 2:** Train weak classifier (e.g., decision stump — 1-level tree)
- **Step 3:** Calculate weighted error rate: $\epsilon = \sum w_i \cdot I(y_i \neq h_i)$
- **Step 4:** Calculate classifier weight: $\alpha = 0.5 \cdot \ln((1-\epsilon)/\epsilon)$
- **Step 5:** Update sample weights — increase for misclassified, decrease for correct
- **Step 6:** Normalize weights so they sum to 1
- **Step 7:** Repeat Steps 2–6 for T iterations
- **Final prediction:** $H(x) = \text{sign}(\sum \alpha_i \cdot h_i(x))$

4.6 Random Forest

- **How:** Builds a large number of decision trees using Bagging + random feature selection at each split (feature subsampling).

- **Feature subsampling:** At each node, only $\sqrt{(\text{total features})}$ features considered for splitting
- **Prediction:** Classification → majority vote; Regression → mean
- **Key advantages:** Robust to outliers, handles missing values, provides feature importance scores, good default performance
- **Key hyperparameters:** `n_estimators` (trees), `max_depth`, `min_samples_split`, `max_features`

4.7 Confusion Matrix & Evaluation Metrics

Term	Definition	Formula
TP (True Positive)	Correctly predicted positive	—
TN (True Negative)	Correctly predicted negative	—
FP (False Positive)	Incorrectly predicted positive (Type I error)	—
FN (False Negative)	Incorrectly predicted negative (Type II error)	—
Accuracy	Overall correctness	$(TP+TN)/(TP+TN+FP+FN)$
Precision	Of predicted positives, how many correct	$TP/(TP+FP)$
Recall (Sensitivity)	Of actual positives, how many caught	$TP/(TP+FN)$
Specificity	Of actual negatives, how many correct	$TN/(TN+FP)$
F1-Score	Harmonic mean of precision & recall	$2 \cdot (P \cdot R) / (P + R)$
ROC-AUC	Area under ROC curve; 1.0 = perfect	TPR vs FPR curve

4.8 Cross Validation

- **K-Fold CV:** Split data into K equal folds. Train on K-1, test on 1. Repeat K times. Final metric = mean ± std of K scores.
- **Stratified K-Fold:** Each fold has same class distribution as full dataset. Recommended for imbalanced datasets.
- **Leave-One-Out (LOO):** K = N. Computationally expensive but low bias.
- **Repeated K-Fold:** K-Fold repeated multiple times with different random splits. More stable estimate.

■ Sklearn: `from sklearn.model_selection import cross_val_score, KFold, StratifiedKFold`

UNIT 5 – Big Data Visualization

5.1 Data Visualization Concepts

Data visualization is the graphical representation of data and information. It uses visual elements like charts, graphs, and maps to provide an accessible way to see and understand trends, outliers, and patterns in data.

- **Why visualize?** Human brain processes images 60,000× faster than text. Visual patterns are immediately apparent.
- **Pre-attentive attributes:** Color, size, shape, position — detected instantly without conscious effort
- **Gestalt principles:** Proximity, similarity, continuity — how humans group visual elements

5.2 Chart Types & When to Use

Chart Type	Best For	Example Use Case
Bar Chart	Comparing categories	Sales by product category
Line Chart	Trends over time	Monthly revenue trend
Pie/Donut	Part-to-whole proportions (<6 categories)	Market share breakdown
Scatter Plot	Correlation between two variables	Age vs income
Histogram	Distribution of continuous variable	Age distribution of customers
Box Plot	Distribution, outliers, quartiles	Salary distribution by department
Heatmap	Matrix/correlation data	Correlation matrix of features
Treemap	Hierarchical part-to-whole	Budget allocation by department
Bubble Chart	3 variables (x, y, size)	Country GDP vs life expectancy vs population

5.3 Visualization Challenges in Big Data

- **Scalability:** Rendering millions of points — use sampling, aggregation, or WebGL-based rendering
- **Interactivity:** Real-time filtering, drill-down, zooming on large datasets
- **Perceptual Limitations:** Wrong chart type misleads — e.g., pie chart with 15 slices
- **Multi-dimensionality:** Visualizing data with >3 dimensions — use PCA, t-SNE, parallel coordinates
- **Real-time Streaming:** Charts must update as data arrives — WebSocket-based solutions
- **Color Blindness:** ~8% of men are color-blind; use colorblind-safe palettes

5.4 Tableau

- **Type:** Market-leading drag-and-drop BI and visualization platform
- **Data sources:** 80+ connectors — Excel, SQL Server, Google Sheets, Salesforce, web data
- **Key concepts:** Dimensions (categorical) vs Measures (numerical), Marks, Shelves, Tooltips
- **Features:** Interactive dashboards, calculated fields, parameters, sets, table calculations
- **Tableau products:** Desktop (local), Server (enterprise), Online (cloud), Public (free, shared)
- **Story Points:** Sequence of visualizations that guide narrative analysis

5.5 D3.js (Data-Driven Documents)

- **Type:** JavaScript library for creating dynamic, interactive web-based data visualizations using SVG, HTML5, CSS
- **Key concept:** Binds data to DOM elements — when data changes, visualization updates automatically
- **Strengths:** Full creative control; custom animations; any chart type imaginable
- **Weaknesses:** Steep learning curve; requires strong JavaScript skills; verbose code
- **Used by:** New York Times, The Guardian, Observable notebook platform

5.6 Google Chart API

- **Type:** Free JavaScript charting library by Google; renders in browser using SVG/Canvas
- **Chart types:** Bar, Line, Pie, Area, Scatter, Geo Map, Gantt, Candlestick, Sankey, Timeline
- **How to use:** Load Google Charts library → create DataTable → specify chart type → draw in <div>
- **Advantages:** Free, no installation, cross-browser, mobile-friendly, easy to embed

5.7 Candela

- **Type:** Open-source visualization component library (Kitware)
- **Built on:** D3.js, Vega, other JS libraries — provides reusable components
- **Key feature:** Works with multiple front-end frameworks (Angular, React, plain JS)
- **Use case:** Scientific data visualization, geospatial data, bioinformatics

UNIT 6 – Big Data Analytics Applications and Tools

6.1 Retail Analytics

- **Customer Segmentation:** K-Means clustering groups customers by purchase behavior → targeted marketing
- **Recommendation Systems:** Collaborative filtering ('Users like you also bought...') — Amazon, Netflix
- **Demand Forecasting:** Time series models (ARIMA, LSTM) predict future demand → reduce inventory costs
- **Price Optimization:** Dynamic pricing based on demand, competition, and customer segments
- **Churn Analysis:** Identify customers likely to stop buying → retention campaigns

6.2 Financial Analytics

- **Fraud Detection:** Real-time anomaly detection on transactions — isolation forest, autoencoders
- **Credit Risk:** Predict loan default probability — logistic regression, gradient boosting
- **Algorithmic Trading:** ML models execute trades in milliseconds based on market signals
- **Portfolio Optimization:** Markowitz mean-variance optimization, reinforcement learning
- **Anti-Money Laundering (AML):** Graph analytics to detect suspicious transaction networks

6.3 Healthcare Analytics

- **Disease Prediction:** Predict patient risk of diabetes, heart disease using ML
- **Medical Imaging:** CNNs analyze X-rays, MRIs for cancer detection (e.g., Google DeepMind eye disease)
- **Drug Discovery:** ML screens millions of molecular candidates faster than traditional methods
- **Hospital Operations:** Predict ER wait times, optimize bed allocation, staff scheduling
- **Readmission Prediction:** Identify high-risk patients before discharge to prevent readmission

6.4 Supply Chain Analytics

- **Demand Forecasting:** Predict future demand to optimize inventory levels
- **Route Optimization:** ML + graph algorithms for optimal delivery routes (UPS ORION system saves 100M miles/year)
- **Supplier Risk:** NLP analysis of news, financials to predict supplier disruptions
- **Warehouse Analytics:** Optimize product placement, picking routes, labor allocation

6.5 HBase

- **Full Name:** Apache HBase — Hadoop Database
- **Type:** Distributed, scalable, column-family NoSQL database running on HDFS
- **Data Model:** Tables → Rows (row key) → Column Families → Columns → Versioned Cell values
- **Key features:** Billions of rows × millions of columns; auto-sharding; strong consistency; random read/write
- **CAP theorem:** CP system (Consistent + Partition tolerant; sacrifices Availability)
- **Use cases:** Facebook Messages (originally), time-series data, IoT sensor data, real-time analytics

6.6 CouchDB

- **Type:** Document-oriented NoSQL database by Apache
- **Format:** Stores and retrieves data as JSON documents via RESTful HTTP API
- **ACID:** ACID compliant at document level (not across documents)
- **Replication:** Master-master replication; excellent for offline-first mobile sync (PouchDB)
- **Views:** MapReduce functions to query and aggregate documents
- **CAP theorem:** AP system (Available + Partition tolerant; eventual consistency)

6.7 Semantria & Mozenda

- **Semantria:** SaaS text analytics and NLP API. Performs sentiment analysis, entity extraction, categorization on social media, reviews, support tickets. Integrates with Excel.
- **Mozenda:** Enterprise web scraping and data extraction platform. Point-and-click agent builder. Outputs to CSV, Excel, databases, APIs. No coding required.

6.8 Big Data Tool Comparison

Tool	Category	Processing Type	Best For
Hadoop (MapReduce)	Distributed processing	Batch	Large-scale ETL, log analysis
Apache Spark	In-memory processing	Batch + Streaming	ML, real-time, iterative jobs
Apache Kafka	Message streaming	Real-time streaming	Event-driven architectures
HBase	NoSQL database	Random read/write	Billions of rows, real-time access
CouchDB	Document database	CRUD + MapReduce	Mobile sync, web apps
Tableau	Visualization/BI	Interactive query	Business dashboards
RapidMiner	ML platform	Visual ML pipeline	No-code ML, predictive analytics
Semantria	Text analytics	NLP API	Sentiment, entity extraction

BUSINESS INTELLIGENCE (BI)

UNIT 1 – Introduction to BI and Decision Support System

1.1 Business Intelligence – Definition & Goals

Business Intelligence (BI) encompasses the strategies, technologies, and tools used by enterprises for data analysis and management of business information. It turns raw data into meaningful, actionable insights.

- **BI Goals:** Improve decision quality, reduce costs, increase revenue, identify market trends, improve operational efficiency
- **BI Process:** Data Collection → Integration → Analysis → Visualization → Decision → Action
- **Key outputs:** Reports, dashboards, KPIs, scorecards, ad-hoc queries, data mining

1.2 BI Architecture & Components

Layer	Component	Purpose
Source Layer	ERP, CRM, OLTP databases, flat files	Original operational data systems
Integration Layer	ETL tools (Informatica, Talend, SSIS)	Extract, clean, and load data
Storage Layer	Data Warehouse, Data Marts	Centralized analytical data store
Analysis Layer	OLAP Engine, Mining tools	Multidimensional analysis, ML
Presentation Layer	Reports, Dashboards, Mobile BI	User-facing visualizations
Governance Layer	Metadata, Security, Data Quality	Ensuring trustworthy, compliant data

1.3 Data → Information → Knowledge → Wisdom (DIKW Hierarchy)

Level	Description	Example
Data	Raw, unprocessed facts and figures	"Sales: 45, 67, 89, 23"
Information	Data processed with context and meaning	"Average monthly sales = 56 units"
Knowledge	Patterns and insights from information	"Sales drop every Q4 due to seasonality"
Wisdom	Applying knowledge to make decisions	"Stock up inventory before Q4 every year"

1.4 Decision Support System (DSS)

A DSS is an interactive information system that supports decision-making by providing data, models, and analysis tools.

- **Components:** Database subsystem, Model base subsystem, User interface subsystem
- **Characteristics:** Semi-structured problems, human judgment involved, flexible and adaptable

DSS Type	Description	Example
Data-driven DSS	Analyzes large databases	Business reporting systems, data warehouses
Model-driven DSS	Uses financial/optimization models	Financial planning, logistics optimization
Communication-driven	Facilitates group decisions	Video conferencing + shared workspace
Document-driven	Retrieves and analyzes documents	Knowledge management systems
Knowledge-driven	Uses expert systems, rules	Medical diagnosis, credit approval

1.5 Simon's Decision-Making Process Model

Stage	Activities	BI Role
Intelligence	Identify problem, gather data, recognize opportunities	Data collection, reports, alerts

Design	Generate alternative solutions, model scenarios	What-if analysis, simulations
Choice	Evaluate alternatives, select best option	Optimization tools, scoring models
Implementation	Execute chosen decision, monitor outcomes	KPI tracking, performance dashboards

1.6 BI–DSS Connection

- BI provides the data infrastructure (DW, ETL, OLAP) that powers DSS analytical capabilities.
- DSS uses BI data and models to support unstructured and semi-structured decisions.
- Modern BI platforms (Power BI, Tableau) embed DSS features: what-if parameters, scenario analysis.

■ *Key exam point: BI = data platform + analysis tools; DSS = decision-focused application built on top of BI data.*

UNIT 2 – Modelling in Business Intelligence

2.1 BI Modelling – Dimensional Modelling

Dimensional modelling organizes data into facts (measurements) and dimensions (context) to optimize analytical query performance.

- **Fact Table:** Contains quantitative measures (sales amount, quantity, profit) and foreign keys to dimension tables. Typically large.
- **Dimension Table:** Contains descriptive attributes (product name, customer region, date). Typically small.
- **Star Schema:** One central fact table directly joined to multiple dimension tables. Simpler queries, faster performance.
- **Snowflake Schema:** Dimensions are normalized into multiple tables. Saves storage, more complex joins.
- **Galaxy Schema:** Multiple fact tables sharing some dimension tables. Also called 'Fact Constellation'.

2.2 Model Building & Assessment

- **Model Building Steps:** Define objectives → Select variables → Choose technique → Train → Validate → Deploy
- **Overfitting:** Model too complex — fits training noise. Solution: regularization, cross-validation, pruning.
- **Underfitting:** Model too simple — misses patterns. Solution: more features, complex model.
- **Model Assessment:** Holdout validation, k-fold CV, ROC-AUC, confusion matrix, lift charts

2.3 Ontology & Frames

- **Ontology:** Formal, explicit specification of concepts, relationships, and constraints within a domain. Used for semantic data integration and knowledge representation.
 - Components: Classes (concepts), Instances (individuals), Properties (attributes/relations), Axioms (rules)
 - Example: Medical ontology (SNOMED CT) — defines diseases, symptoms, treatments and their relationships
- **Frames:** Data structures (like objects in OOP) representing stereotyped situations with slots (attributes) and slot values. Precursor to object-oriented programming and ontologies.

2.4 BPMN (Business Process Model and Notation)

BPMN is the standard graphical notation for documenting and communicating business processes. ISO/IEC 19510 standard.

BPMN Element	Shape	Description
Start Event	Thin circle	Triggers the process (message, timer, signal)
End Event	Thick circle	Terminates the process
Task/Activity	Rounded rectangle	Unit of work performed by human or system
Gateway	Diamond	Decision point (XOR=exclusive, AND=parallel, OR=inclusive)
Sequence Flow	Solid arrow	Order of activities in the process
Message Flow	Dashed arrow	Communication between two pools (organizations)
Pool/Lane	Rectangle with label	Participant or role in the process

2.5 Probabilistic Structures

- **Bayesian Networks:** Directed Acyclic Graph (DAG) representing probabilistic dependencies between variables. Each node = variable; edge = conditional dependency.
 - Used for: diagnosis (medicine), risk analysis, spam filtering
- **Markov Chains:** Stochastic model where future state depends ONLY on current state (memoryless / Markov property).

■ Used for: weather prediction, text generation, PageRank

- **Hidden Markov Models (HMM):** States are hidden (not directly observable). Used for speech recognition, sequence labeling.

2.6 Analytical Structures – OLAP

- **OLAP Cube:** Multidimensional data structure allowing fast analysis across multiple dimensions (e.g., Product × Region × Time).
- **ROLAP:** Relational OLAP — uses relational database with star schema. Scalable.
- **MOLAP:** Multidimensional OLAP — data pre-aggregated in cube. Fastest queries.
- **HOLAP:** Hybrid OLAP — combines ROLAP + MOLAP.

OLAP Operation	Description	Example
Roll-Up (Drill-Up)	Aggregate data to higher level	Daily sales → Monthly → Yearly
Drill-Down	Navigate to more detailed level	Country → State → City → Store
Slice	Select one dimension value (2D view)	Sales for Year=2023 only
Dice	Select subset of multiple dimensions	Sales in East region, Q1, for Electronics
Pivot (Rotate)	Rotate cube to view from different angle	Rows → Columns swap

2.7 Data Quality Dimensions

Dimension	Description	How to Measure
Accuracy	Data correctly represents reality	Compare to verified source
Completeness	No missing values	% non-null values
Consistency	Same data across systems	Cross-system comparison
Timeliness	Data is current and available when needed	Age of data vs freshness requirement
Validity	Data conforms to defined formats/rules	Regex checks, domain validation
Uniqueness	No duplicates exist	Duplicate record count

2.8 Role of Time in BI

- **Time Dimension:** Almost every BI system has a Date dimension with hierarchy: Day → Week → Month → Quarter → Year
- **Slowly Changing Dimensions (SCD):** Handle changes in dimension attributes over time.
 - SCD Type 1: Overwrite old value — no history kept
 - SCD Type 2: Add new row with effective date range — full history kept (most common)
 - SCD Type 3: Add new column for previous value — limited history
- **Temporal Analytics:** Year-over-year (YoY), Month-to-date (MTD), Rolling averages, Seasonality analysis

UNIT 3 – Data Provisioning and Data Visualization

3.1 Data Warehouse – Deep Dive

A Data Warehouse (DW) is a subject-oriented, integrated, non-volatile, and time-variant (SINT) collection of data that supports management's decision-making process. (W.H. Inmon, 1990)

Characteristic	Meaning
Subject-Oriented	Organized around major subjects (Sales, HR, Finance) not business transactions
Integrated	Data from disparate sources unified with consistent naming, formats, units
Non-Volatile	Data is not updated/deleted — only appended (history preserved)
Time-Variant	Data has time dimension — changes tracked over time periods

- **OLTP vs OLAP:** OLTP (operational DB) — fast CRUD, current data; OLAP (DW) — complex queries, historical data, read-optimized

3.2 ETL Process – Detailed

- **Extract:** Pull data from OLTP databases, flat files, APIs, SaaS applications, web services
 - Full extraction: all data every time (simple but slow)
 - Incremental extraction: only changed records since last run (efficient)
- **Transform:** Apply business rules and data quality operations:
 - • Cleansing: fix typos, standardize formats, handle nulls
 - • Integration: resolve naming conflicts across systems
 - • Aggregation: pre-compute summaries for performance
 - • Derivation: calculate new fields (e.g., profit = revenue – cost)
 - • Encoding: convert categorical to numeric, normalize dates
- **Load:** Insert transformed data into target (DW, data mart, data lake)
 - Full load: replace all data (simple, works for small tables)
 - Incremental load: insert/update only new/changed records (efficient for large tables)
- **ETL Tools:** Informatica PowerCenter, Microsoft SSIS, Talend, Apache Spark, AWS Glue, dbt

3.3 Data Profiling & Enrichment

- **Data Profiling:** Systematic analysis of data to assess its quality, structure, and content before ETL.
 - Column profiling: data types, null %, distinct values, min/max, patterns
 - Cross-column profiling: functional dependencies, correlations between columns
 - Cross-table profiling: referential integrity, join quality
- **Data Enrichment:** Adding external data to enhance existing records:
 - Example 1: Append zip code → demographics (income, population)
 - Example 2: Append IP address → geolocation data
 - Example 3: Append company name → firmographic data (size, industry)

3.4 Data Marts & OLAP Cubes

- **Data Mart:** Subject-specific subset of a data warehouse (e.g., Sales Data Mart, HR Data Mart)
 - Dependent: derives data from DW — consistent, but DW must exist first
 - Independent: gets data directly from source — faster to build, may be inconsistent
- **OLAP Cube:** Pre-aggregated multidimensional dataset stored for fast analytical queries. Dimensions = axes of analysis; Measures = facts being analyzed.

3.5 Business Reports & Dashboards

- **Operational Reports:** Detailed, frequently generated, specific audience (e.g., daily inventory report)
- **Analytical Reports:** Trend analysis, comparative, exploratory (e.g., quarterly performance review)
- **Dashboards:** Real-time visual display of KPIs on a single screen. Three types:
 - Strategic dashboards: high-level KPIs for executives
 - Analytical dashboards: trends and patterns for analysts
 - Operational dashboards: real-time operational metrics
- **Scorecards:** Compare actual KPIs against targets — show performance as %, RAG (Red/Amber/Green) status

3.6 Power BI

- **Architecture:** Power Query (ETL) → Data Model (DAX) → Visualizations → Reports → Dashboards → Sharing
- **Power Query:** Built-in ETL tool. Connects to 100+ sources; transforms data using M language.
- **Data Model:** Star schema relationships between tables. Supports calculated columns and measures.
- **DAX (Data Analysis Expressions):** Formula language for calculated fields:
 - Measures: CALCULATE, SUM, AVERAGE, COUNTROWS, DIVIDE, ALL, FILTER
 - Time intelligence: TOTALYTD, SAMEPERIODLASTYEAR, DATESYTD
- **Visualizations:** Bar, line, pie, matrix, card, map, decomposition tree, AI visuals, custom visuals

3.7 Tableau (BI Context)

- **Tableau Prep:** ETL tool for data preparation and cleaning before visualization
- **LOD Expressions:** Level of Detail — {FIXED [Region] : SUM([Sales])} — compute at different granularity than view
- **Parameters:** Dynamic variables users can control in dashboards
- **Tableau Server/Online:** Publish and share dashboards with organization; manage permissions, schedules

3.8 Oracle BI (OBIEE)

- **OBIEE:** Oracle Business Intelligence Enterprise Edition — enterprise-grade BI platform
- **Repository (RPD):** Metadata model with three layers:
 - Physical Layer: actual database tables, connections, joins
 - Business Model & Mapping Layer: logical facts, dimensions, metrics
 - Presentation Layer: what users see in reports — folder structure
- **Answers:** Ad-hoc query and report builder
- **Dashboards:** Assemble reports into interactive pages

UNIT 4 – Data Pre-processing Techniques

4.1 Data Validation

- **Syntax validation:** Check format (e.g., email must contain @, date must be YYYY-MM-DD)
- **Range/domain validation:** Age must be 0–150; temperature in valid range
- **Referential integrity:** Foreign key values must exist in parent table
- **Business rule validation:** End date must be > Start date; quantity must be positive
- **Cross-field validation:** If marital_status='Single', num_children should typically be low

4.2 Noise Handling

Noise is random error or variance in a measured variable. Sources include faulty sensors, human entry errors, or transmission errors.

- **Binning (Smoothing):** Sort values → divide into bins → replace each value with bin mean/median/boundary
 - Equal-width bins: each bin has same range of values
 - Equal-frequency bins: each bin has same number of values
- **Regression smoothing:** Fit data to regression curve; use predicted values as smoothed data
- **Clustering:** Values that don't belong to any cluster are treated as outliers/noise

4.3 Standardization & Normalization

Method	Formula	Result Range	When to Use
Min-Max Normalization	$x' = (x - \min) / (\max - \min)$	[0, 1]	KNN, neural networks (fixed range needed)
Z-score Standardization	$x' = (x - \mu) / \sigma$	~[-3, +3]	SVM, linear regression (assumes normal distribution)
Decimal Scaling	$x' = x / 10^j$ (where j makes $ x' \leq 1$)	It varies	Simple normalization for whole numbers
Robust Scaling	$x' = (x - \text{median}) / \text{IQR}$	Robust to outliers	Data with outliers

4.4 Feature Extraction

- **Definition:** Transform raw data into a set of engineered features that better represent the underlying problem to the predictive model.
- **Text:** TF-IDF, word embeddings (Word2Vec), bag-of-words
- **Images:** HOG features, CNN activations as features
- **Time Series:** Rolling mean, lag features, FFT components
- **PCA:** Linear transformation that finds directions of maximum variance (principal components)

4.5 Feature Selection

Feature selection reduces dimensionality by choosing the most informative features, reducing overfitting and improving model speed.

Method	Technique	Pros / Cons
Filter Methods	Pearson correlation, Chi-square, Information Gain, ANOVA F-test	Fast and simple, but ignores feature interactions
Wrapper Methods	Forward selection, Backward elimination, Best-First search, etc.	Considers feature interactions, but computationally expensive
Embedded Methods	LASSO (L1 reg.), Ridge (L2 reg.), Decision Trees, etc.	Feature selection and regularization / Model-specific

4.6 PCA (Principal Component Analysis) – Step by Step

- **Step 1:** Standardize data (zero mean, unit variance)
- **Step 2:** Compute covariance matrix $C = (1/n) X^T X$
- **Step 3:** Compute eigenvectors and eigenvalues of C

- **Step 4:** Sort eigenvectors by eigenvalue (descending)
- **Step 5:** Select top K eigenvectors (principal components) that explain $\geq 95\%$ variance
- **Step 6:** Project data: $Z = X \cdot W$ (W = matrix of top K eigenvectors)
- **Output:** K-dimensional representation of original N-dimensional data

4.7 Sampling Methods

Method	How	Best For
Simple Random	Each sample equally likely to be selected	Homogeneous populations
Systematic	Select every K-th item from ordered list	Large, ordered datasets
Stratified	Sample proportionally from predefined subgroups	Heterogeneous populations, imbalanced classes
Cluster	Randomly select clusters; sample all in selected clusters	Geographically spread population
Convenience	Use whatever data is easily available	Quick estimates (biased)

4.8 Data Exploration – Univariate/Bivariate/Multivariate

- **Univariate:** One variable. Statistics: mean, median, mode, std dev, skewness, kurtosis. Charts: histogram, box plot, bar chart, KDE plot.
- **Bivariate:** Two variables. Statistics: correlation (Pearson r for continuous, Spearman ρ for ordinal), cross-tabulation (categorical). Charts: scatter plot, grouped bar, line chart.
- **Multivariate:** Multiple variables. Techniques: PCA, factor analysis, multiple regression. Charts: heatmap (correlation matrix), pair plot, parallel coordinates, 3D scatter.

■ *Skewness: positive = tail on right; negative = tail on left. Kurtosis: high = heavy tails (more outliers).*

UNIT 5 – Impact of Machine Learning in BI

5.1 ML in BI – Overview

Machine Learning enhances BI by automating pattern discovery, enabling predictive analytics, personalizing insights, and reducing reliance on manual analysis. The combination creates 'Augmented Analytics'.

Traditional BI	ML-Enhanced BI
Answers "what happened"	Answers "what will happen" and "what should we do"
Requires analyst to ask right questions	Automatically surfaces anomalies and insights
Historical, descriptive reports	Predictive and prescriptive recommendations
Manual threshold alerts	ML-powered anomaly detection

5.2 Regression in BI

- **Linear Regression:** Predict continuous outcomes — sales forecasting, demand planning, price prediction
- **Multiple Regression:** Multiple predictors — customer lifetime value = $f(\text{recency, frequency, monetary})$
- **Time Series Regression:** Capture temporal patterns — ARIMA, Exponential Smoothing, Prophet (Facebook)
- **Polynomial Regression:** Non-linear relationships — $y = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$

5.3 Classification Algorithms in BI

Algorithm	How it Works	BI Application
Decision Tree	Recursively splits data by most informative feature	Customer Segmentation, loan approval
Random Forest	Ensemble of decision trees (bagging + random features)	Fraud detection, churn prediction
Naive Bayes	Bayes theorem with feature independence assumption	Spam classification, sentiment analysis
SVM	Find optimal hyperplane maximizing margin between classes	Text classification, image recognition
KNN	Classify by majority vote of K nearest training points	Recommendation, anomaly detection
Neural Networks	Layers of neurons learning non-linear patterns	Image recognition, NLP, complex patterns

5.4 Bayesian Methods

Bayesian methods use Bayes' theorem to update probability estimates as new data becomes available.

- **Bayes' Theorem:** $P(A|B) = P(B|A) \times P(A) / P(B)$
 - $P(A)$ = Prior probability; $P(B|A)$ = Likelihood; $P(A|B)$ = Posterior probability; $P(B)$ = Evidence
- **Naive Bayes Classifier:** Assumes all features are conditionally independent given the class.
 - $P(\text{Class}|\text{features}) \propto P(\text{Class}) \times \prod P(\text{feature}_i|\text{Class})$
- **Variants:** Gaussian NB (continuous), Multinomial NB (text counts), Bernoulli NB (binary features)
- **BI Applications:** Email spam filtering, document categorization, medical diagnosis, fraud scoring

5.5 Clustering in BI

- **K-Means:** Customer segmentation (RFM analysis: Recency, Frequency, Monetary), market segmentation
- **Hierarchical Clustering:** No need to specify K; dendrogram shows all possible clusterings; good for small datasets
 - Agglomerative (bottom-up): each point starts as own cluster → merge closest pairs → repeat
 - Divisive (top-down): all points in one cluster → split recursively
 - Linkage: Single (min distance), Complete (max distance), Average, Ward (min variance)
- **DBSCAN:** Density-Based Spatial Clustering. Finds clusters of arbitrary shape, marks outliers as noise.
 - Parameters: ϵ (neighborhood radius), MinPts (minimum points to form dense region)

5.6 Association Rules in BI

- **Market Basket Analysis:** Discover products frequently purchased together → cross-sell, store layout
- **Apriori Algorithm:** Find frequent itemsets above min_support threshold, generate rules above min_confidence
- **FP-Growth Algorithm:** Faster than Apriori — uses FP-tree structure, no candidate generation
- **BI Applications:** Retail cross-selling, website navigation patterns, medical symptom co-occurrence

■ *Example: IF {diapers} THEN {beer} — famous example. Lift > 1 indicates real association, not random co-occurrence.*

UNIT 6 – BI Applications, Emerging Trends and Future Impacts

6.1 BI in Healthcare

- **Clinical Analytics:** Predict patient readmission risk (CMS penalty avoidance), length-of-stay prediction
- **Population Health Management:** Track chronic disease prevalence, vaccination coverage, disease outbreak detection
- **Operational Analytics:** Optimize bed utilization, OR scheduling, supply chain for pharmaceuticals
- **Regulatory Compliance:** HIPAA compliance reporting, quality measure reporting (HEDIS, STARS)
- **Real-world example:** Kaiser Permanente uses predictive analytics to identify high-risk patients before crisis

6.2 BI in Education

- **Learning Analytics:** Track student engagement, assignment completion, predict final grades
- **Early Warning Systems:** Identify at-risk students early using attendance, grades, behavior data
- **Curriculum Optimization:** Analyze assessment data to improve teaching content
- **Institutional BI:** Enrollment forecasting, faculty workload analysis, financial sustainability
- **Adaptive Learning:** Personalized learning paths based on individual student performance data

6.3 BI in CRM

- **Customer 360 View:** Unified profile combining all customer interactions across channels
- **Customer Segmentation:** RFM analysis, psychographic and behavioral segments → personalized campaigns
- **Churn Prediction:** ML model identifies customers likely to leave → proactive retention
- **Sentiment Analysis:** NLP on reviews, social media, support tickets → understand customer satisfaction
- **Next Best Action:** AI recommends optimal next interaction for each customer

6.4 BI in Banking, Telecom & Manufacturing

- **Banking:** Credit scoring (FICO), AML (anti-money laundering), regulatory capital calculations, real-time fraud alerts
- **Telecom:** Network performance analytics, subscriber churn prediction, revenue assurance, tower planning
- **Manufacturing:** Predictive maintenance (predict machine failure before it occurs), quality control, OEE (Overall Equipment Effectiveness) dashboards, supply chain optimization

6.5 Mobile BI

Mobile BI enables access to business intelligence, reports, and dashboards on smartphones and tablets.

- **Key features:** Responsive/adaptive design, touch-optimized charts, offline access, push notifications for KPI alerts
- **Challenges:** Small screen real estate, security (MDM, MFA), data connectivity, battery constraints
- **Tools:** Power BI Mobile (iOS/Android), Tableau Mobile, MicroStrategy Mobile, SAP Analytics Cloud Mobile
- **Design principles:** Simplify to 3–5 key metrics, use large touch targets, prioritize speed over complexity

6.6 Cloud Computing & BI

- **Cloud BI benefits:** No infrastructure investment, instant scalability, pay-as-you-go, global accessibility
- **Deployment models:** SaaS BI (Power BI Online, Tableau Cloud), PaaS (AWS QuickSight), IaaS (self-managed on cloud VMs)
- **Major platforms:** AWS (QuickSight, Redshift), Azure (Synapse, Power BI), GCP (Looker, BigQuery)
- **Data Lake:** Store raw data in native format at scale → analyze with Spark, query with Athena/BigQuery
- **Challenges:** Data sovereignty & compliance, latency for real-time, vendor lock-in, data egress costs

6.7 Social Networking Analytics

- **Sentiment Analysis:** Classify social media posts as positive/negative/neutral using NLP
- **Trend Detection:** Identify emerging topics using TF-IDF, topic modeling (LDA), hashtag analysis
- **Influencer Identification:** Graph centrality measures (degree, betweenness, PageRank) to find key nodes
- **Community Detection:** Graph clustering algorithms to find communities/clusters in social networks
- **Tools:** Semantria, Brandwatch, Hootsuite Insights, Sprinklr, Crimson Hexagon

6.8 Emerging Trends in BI

- **Augmented Analytics:** AI/ML automates data preparation, insight generation, and narrative explanations. Users ask questions in natural language.
- **Natural Language Processing (NLP):** Query data using English sentences: 'Show me top 5 products by sales last month'. Power BI Q&A, Tableau Ask Data.
- **Self-Service BI:** Business users create own reports without IT help. Democratizes data access.
- **Embedded BI:** BI functionality embedded directly into business applications (CRM, ERP) — users don't leave their workflow.
- **Real-time & Streaming BI:** Process and visualize data as it arrives — IoT dashboards, live trading screens, logistics tracking.
- **AI-Powered Anomaly Detection:** Automatically detect unusual patterns and alert analysts. No need to define static thresholds.
- **Data Fabric & Data Mesh:** Modern architectures for distributed data ownership and access across the organization.

QUICK REVISION CHEAT SHEET

All Important Formulas

Formula Name	Equation	Subject Area
Euclidean Distance	$d = \sqrt{\sum (x_i - y_i)^2}$	K-Means, KNN
WCSS (Elbow Method)	$\sum \sum x_i - c_k ^2$	K-Means validation
Silhouette Score	$s(i) = (b(i) - a(i)) / \max(a(i), b(i))$	Cluster validation
Support (Apriori)	$\text{freq}(A \cup B) / N$	Association Rules
Confidence	$\text{Support}(A \cup B) / \text{Support}(A)$	Association Rules
Lift	$\text{Confidence}(A \rightarrow B) / \text{Support}(B)$	Association Rules
Simple Linear Regression	$y = b_0 + b_1x$	Regression
Multiple Regression	$y = b_0 + b_1x_1 + \dots + b_nx_n$	Regression
Sigmoid (Logistic)	$\sigma(z) = 1 / (1 + e^{-z})$	Logistic Regression
Bayes' Theorem	$P(A B) = P(B A) \cdot P(A) / P(B)$	Naive Bayes, Probabilistic Models
Accuracy	$(TP+TN) / (TP+TN+FP+FN)$	Model Evaluation
Precision	$TP / (TP+FP)$	Model Evaluation
Recall (Sensitivity)	$TP / (TP+FN)$	Model Evaluation
Specificity	$TN / (TN+FP)$	Model Evaluation
F1-Score	$2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$	Model Evaluation
Min-Max Normalization	$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$	Pre-processing
Z-score Standardization	$x' = (x - \mu) / \sigma$	Pre-processing
AdaBoost Classifier Weight	$\alpha_i = 0.5 \cdot \ln((1 - \epsilon_i) / \epsilon_i)$	AdaBoost
MSE (Regression error)	$\text{MSE} = (1/n) \sum (y_i - \hat{y}_i)^2$	Regression evaluation

Key Abbreviations Quick Reference

Abbr.	Full Form	Abbr.	Full Form
HDFS	Hadoop Distributed File System	OLAP	Online Analytical Processing
YARN	Yet Another Resource Negotiator	OLTP	Online Transaction Processing
ETL	Extract, Transform, Load	DW	Data Warehouse
EDA	Exploratory Data Analysis	DSS	Decision Support System
PCA	Principal Component Analysis	KPI	Key Performance Indicator
BPMN	Business Process Model and Notation	SCD	Slowly Changing Dimension
AUC	Area Under the Curve	ROC	Receiver Operating Characteristic
RFM	Recency, Frequency, Monetary	LDA	Latent Dirichlet Allocation
BASE	Basically Available, Soft state, Eventual Consistency	ACID	Atomicity, Consistency, Isolation, Durability
BI	Business Intelligence	ML	Machine Learning
NLP	Natural Language Processing	OEE	Overall Equipment Effectiveness
WCSS	Within-Cluster Sum of Squares	HMM	Hidden Markov Model
SVM	Support Vector Machine	KNN	K-Nearest Neighbours
GMM	Gaussian Mixture Model	RFE	Recursive Feature Elimination

OBIEE	Oracle BI Enterprise Edition	RPD	Repository (Oracle BI)
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